

Proof of Concept Report

Splunk SSH Failed Login Detection

1. Project Overview

This proof of concept demonstrates a small SIEM detection lab using Splunk Enterprise on Kali Linux. The lab uses synthetic Linux SSH authentication events to identify repeated failed login attempts and present the results through an alert and dashboard.

2. Objectives

The objectives were to install and run Splunk, ingest mock authentication logs, write an SPL detection for repeated failed SSH logins, create an alert, and build a dashboard for monitoring attacking source IPs.

3. Environment

Operating system: Kali Linux. SIEM platform: Splunk Enterprise 10.4.4. Data source: synthetic SSH authentication log events. Detection language: Splunk Search Processing Language (SPL).

4. Detection Query

The primary SPL detection searches for events containing 'Failed password', extracts the source IP address, counts events by source IP, filters for more than five failures, and sorts the results in descending order.

5. SPL Query

```
index=main "Failed password"
| rex "from (?<src_ip>\d{1,3}(?:\.\d{1,3}){3})"
| stats count as failed_attempts by src_ip
| where failed_attempts > 5
| sort - failed_attempts
```

6. Results

The lab dashboard showed three source IPs exceeding the configured threshold: 10.10.10.25 with 18 failed attempts, 10.10.10.31 with 11 failed attempts, and 10.10.10.44 with 7 failed attempts. These are synthetic/private addresses used only for the lab.

7. Alert

A Splunk alert was configured around the failed-login detection. Its purpose is to identify source IPs that exceed the configured threshold and provide an actionable signal for further investigation.

8. Dashboard

The dashboard is named 'SSH Authentication Threat Monitor'. It contains a 'Failed Attempts by IP' visualization and a 'Top Attacking IPs' table, allowing the analyst to quickly identify high-volume failed-login sources.

9. Security Interpretation

Repeated failed SSH authentication can be consistent with password guessing or brute-force behavior. The detection is intended as an initial signal rather than proof of compromise. An analyst should review the time period, targeted accounts, source context, and any successful authentication events before making a final

determination.

10. Limitations

The lab uses synthetic data and a simple threshold. The more-than-five rule is a demonstration threshold, not a universal production standard. Real deployments should use appropriate time windows, tuning, allowlists, account context, source reputation, and false-positive analysis.

11. Evidence Note

Available evidence includes setup, SPL command, alert, and dashboard screenshots. The separate output screenshot was not captured after a system restart because the Splunk service required manual startup near the submission deadline. The dashboard preserves the demonstrated detection results.

12. Conclusion

The proof of concept demonstrates a complete basic SIEM workflow: authentication log ingestion, SPL detection, alerting, and dashboard visualization. It provides a foundation for future improvements such as time-windowed brute-force detection, password-spraying detection, and failed-then-successful login correlation.

Observed Detection Results

Source IP	Failed Attempts
10.10.10.25	18
10.10.10.31	11
10.10.10.44	7